

Amrith Akshintala

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GitHub: <https://github.com/CodingMastermind123> **Portfolio:** [https://codingmastermind123.github.io/Main-](https://codingmastermind123.github.io/Main-Portfolio/)

[Portfolio/](#)

OBJECTIVE

Rising Sophomore Engineering Honors student at Texas A&M with hands-on experience in embedded systems, robotics, and software development through project teams and research. Eager to develop practical autonomous systems and real-time software development with mentorship from experienced engineers.

EDUCATION

Texas A&M University — College Station, TX

Graduation: May 2029

- B.S. Computer Engineering | GPA: 3.80 | Engineering Honors

TECHNICAL PROJECTS

Sensor Dashboard — Personal Project

July 2026 - Present

- Built a full-stack real-time telemetry dashboard for embedded sensor data, with a Node.js/serialport bridge streaming Arduino Uno R4 data through an Express + WebSocket backend to a React/Vite frontend
- Implemented live charting with Recharts, a drag-and-resize dashboard layout using react-grid-layout, and CSV session logging via a custom recorder module

2-Axis Gimbal Project - Personal Project

May 2026 – Present

- Designing a 2-axis camera gimbal with a team of 4 using Arduino GY-87 IMU and SG90 servos; implemented PID stabilization using C++
- Completed full CAD modeling and assembly in SolidWorks; currently in final wiring and hardware assembly

StudyBot — Personal Project

June 2026 – Present

- Built and deployed a full-stack AI study assistant (React, Node/Express, Supabase, Claude API) with Google OAuth, row-level security, prompt caching (~90% cost reduction), and an interactive quiz system
- Engineered a RAG pipeline with semantic vector search (OpenAI embeddings + Supabase pgvector)

Disease Classification from Medical Images — Aggie Data Science Club

Jan 2026 – April 2026

- Built a ResNet-50 deep learning model to classify chest X-rays into 3 categories (Normal, Bacterial Pneumonia, Viral Pneumonia), achieving 86.4% validation accuracy
- Performed data preprocessing, augmentation, and evaluation using PyTorch, NumPy, and Pandas

Cosmos — Hackathon

April 2026

- Built an interactive force-directed knowledge graph visualizing NASA exoplanet and mission data, with an AI guide powered by the Claude API; Developed in React and D3.js; presented at Build4Good hackathon

RELEVANT EXPERIENCE

Member—Texas A&M TURTLE Robotics

Sep 2025 – Dec 2025

- Developed foundational skills in Python, C++, electronics, CAD modeling, circuits, and manufacturing
- Collaborated with a team to design, build, and test a robot a block-retrieval challenge against opposing robots; designed all components in SolidWorks

Research Intern — Texas A&M Ultrasound Research Laboratory

Aug 2025 – Present

- Developed non-invasive ultrasound techniques to measure tumor pressure

Research Intern — Dartmouth-Hitchcock Medical Center

Jun 2024 – Aug 2024

- Conducted biomedical research using computational models for disease diagnosis and pathological data analysis

VOLUNTEER EXPERIENCE

Co-founder — Scholar Sphere

Jul 2023 – Jul 2025

- Started a nonprofit and developed a website providing online educational resources to high school students
- Developed an online SAT practice platform with full-length test simulation using HTML, CSS, and JavaScript
- Established partnerships with 22 nonprofits; earned the PVSA Gold Award for volunteer service

Lead Tutor — Voices Academy

Nov 2021 – Dec 2023

- Led virtual tutoring sessions for K–6 students from low-income backgrounds

SKILLS

Technical: Python, Java, JavaScript, HTML, CSS, React, Swift, SystemVerilog, TensorFlow, PyTorch, SolidWorks, Research, Lab techniques

Professional: Leadership, Teamwork, Time Management, Problem-Solving, Communication

AWARDS & HONORS

Eagle Scout | AP Scholar with Distinction | National Merit Commended Scholar | PVSA Gold Award